

$$\begin{aligned} & (g_1^2 + g_2^2) + \\ & \frac{1}{16\pi^2} \left(\frac{1}{24} (-3g_1^4 - 5g_2^4) + \frac{4}{9} C_{\phi 2^2} g_2^4 LF_{3,0}[m_2] + \frac{2}{3} C_{\phi 2^2} g_2^4 LF_{4,-1}[m_2] - \frac{32}{45} C_{\phi 2^2} g_2^4 LF_{5,-2}[m_2] - \right. \\ & \frac{1}{12} \sum_p g_1^4 LF_{2,0}[m_d^p] + g_1^2 \overline{y_d^{pr}} y_d^{pr} LF_{2,0}[m_d^r] - \frac{1}{4} \sum_p g_1^4 LF_{2,0}[m_e^p] + \\ & g_1^2 \overline{y_e^{pr}} y_e^{pr} LF_{2,0}[m_e^r] + \frac{1}{8} (4 \overline{y_e^{pr}} y_e^{pr} (-g_1^2 + g_2^2) - \sum_p (g_1^4 + g_2^4)) LF_{2,0}[m_l^p] + \frac{1}{18} C_{\phi 2^2} \\ & g_2^2 (36 \overline{y_e^{pr}} y_e^{pr} - 5 \sum_p g_2^2) LF_{3,0}[m_l^p] + \frac{1}{12} C_{\phi 2^2} g_2^2 (-24 \overline{y_e^{pr}} y_e^{pr} + \sum_p g_2^2) LF_{4,-1}[m_l^p] + \\ & \frac{8}{45} \sum_p C_{\phi 2^2} g_2^4 LF_{5,-2}[m_l^p] + \left(\frac{1}{2} \overline{y_d^{pr}} y_d^{pr} (g_1^2 + 3g_2^2) - \frac{1}{24} \sum_p (g_1^4 + 9g_2^4) \right) LF_{2,0}[m_q^p] + \\ & \frac{1}{6} C_{\phi 2^2} g_2^2 (36 \overline{y_d^{pr}} y_d^{pr} - 5 \sum_p g_2^2) LF_{3,0}[m_q^p] + \frac{1}{4} C_{\phi 2^2} g_2^2 (-24 \overline{y_d^{pr}} y_d^{pr} + \sum_p g_2^2) LF_{4,-1}[m_q^p] + \\ & \frac{8}{15} \sum_p C_{\phi 2^2} g_2^4 LF_{5,-2}[m_q^p] - \frac{1}{3} \sum_p g_1^4 LF_{2,0}[m_u^p] + \frac{2}{9} C_{\phi 2^2} g_2^4 LF_{3,0}[\tilde{\mu}] + \\ & \frac{1}{3} C_{\phi 2^2} g_2^4 LF_{4,-1}[\tilde{\mu}] - \frac{16}{45} C_{\phi 2^2} g_2^4 LF_{5,-2}[\tilde{\mu}] + \frac{1}{2} g_1^4 LF_{2,2,-2}[m_1, \tilde{\mu}] + \\ & \frac{1}{2} g_1^4 m_1^2 LF_{2,2,-1}[m_1, \tilde{\mu}] - \frac{8}{3} C_{\phi 2^2} g_2^4 LF_{2,1,0}[m_2, \tilde{\mu}] + \frac{5}{2} g_2^4 LF_{2,2,-2}[m_2, \tilde{\mu}] + \\ & \frac{1}{6} g_2^4 (-34 C_{\phi 2^2} + 3m_2^2) LF_{2,2,-1}[m_2, \tilde{\mu}] + \frac{2}{3} m_2 \tilde{\mu} g_2^4 (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) LF_{2,2,0}[m_2, \tilde{\mu}] + \\ & \frac{4}{3} C_{\phi 2^2} g_2^4 LF_{3,1,-1}[m_2, \tilde{\mu}] + \frac{28}{3} C_{\phi 2^2} g_2^4 LF_{3,2,-2}[m_2, \tilde{\mu}] - \\ & \frac{1}{3} m_2 g_2^4 (18 m_2 C_{\phi 2^2} + \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1})) LF_{3,2,-1}[m_2, \tilde{\mu}] - 4 C_{\phi 2^2} g_2^4 LF_{3,3,-3}[m_2, \tilde{\mu}] + \\ & 4 C_{\phi 2^2} g_2^4 m_2^2 LF_{3,3,-2}[m_2, \tilde{\mu}] - 4 C_{\phi 2^2} g_2^4 LF_{4,2,-3}[m_2, \tilde{\mu}] + 4 C_{\phi 2^2} g_2^4 m_2^2 LF_{4,2,-2}[m_2, \tilde{\mu}] - \\ & 3 \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{pt} y_d^{sr} LF_{1,1,0}[m_d^r, m_d^t] + g_1^2 \overline{a_d^{pr}} a_d^{pr} LF_{2,1,0}[m_d^r, m_q^p] + \\ & \frac{1}{2} g_1^2 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} + \overline{a_d^{pr}} (2 C_{\phi 2^2} a_d^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_d^{pr})) LF_{2,2,0}[m_d^r, m_q^p] - \\ & \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{pt} y_e^{sr} LF_{1,1,0}[m_e^r, m_e^t] + g_1^2 \overline{a_e^{pr}} a_e^{pr} LF_{2,1,0}[m_e^r, m_l^p] + \\ & \frac{1}{2} g_1^2 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} + \overline{a_e^{pr}} (2 C_{\phi 2^2} a_e^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_e^{pr})) LF_{2,2,0}[m_e^r, m_l^p] + \\ & g_2^2 \overline{a_e^{pr}} a_e^{pr} LF_{2,1,0}[m_l^p, m_e^r] - \frac{1}{2} \overline{a_e^{pr}} a_e^{pr} (g_1^2 + g_2^2) LF_{3,1,-1}[m_l^p, m_e^r] + \\ & g_2^2 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} + \overline{a_e^{pr}} (2 C_{\phi 2^2} a_e^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_e^{pr})) LF_{3,1,0}[m_l^p, m_e^r] - \\ & \frac{1}{2} g_1^2 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} + \overline{a_e^{pr}} (2 C_{\phi 2^2} a_e^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_e^{pr})) LF_{3,2,-1}[m_l^p, m_e^r] + \\ & \frac{1}{4} (-2 C_{\phi 2^2} \overline{a_e^{pr}} a_e^{pr} (3g_1^2 + 5g_2^2) - \tilde{\mu} (3g_1^2 + 7g_2^2) (C_{\phi 2\phi 1} \overline{y_e^{pr}} a_e^{pr} + \overline{C_{\phi 2\phi 1}} y_e^{pr} \overline{a_e^{pr}})) \\ & LF_{4,1,-1}[m_l^p, m_e^r] + \\ & \frac{1}{3} (2 C_{\phi 2^2} \overline{a_e^{pr}} a_e^{pr} (3g_1^2 + g_2^2) + \tilde{\mu} (3g_1^2 + 2g_2^2) (C_{\phi 2\phi 1} \overline{y_e^{pr}} a_e^{pr} + \overline{C_{\phi 2\phi 1}} y_e^{pr} \overline{a_e^{pr}})) \\ & LF_{5,1,-2}[m_l^p, m_e^r] - \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{pt} y_e^{sr} LF_{1,1,0}[m_l^p, m_l^s] - \\ & 2 C_{\phi 2^2} \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{pt} y_e^{sr} LF_{2,1,0}[m_l^p, m_l^s] + 2 C_{\phi 2^2} \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{pt} y_e^{sr} LF_{3,1,-1}[m_l^p, m_l^s] + \\ & \overline{a_d^{pr}} a_d^{pr} (2g_1^2 + 3g_2^2) LF_{2,1,0}[m_q^p, m_d^r] - \frac{3}{2} \overline{a_d^{pr}} a_d^{pr} (g_1^2 + g_2^2) LF_{3,1,-1}[m_q^p, m_d^r] + \\ & (2g_1^2 + 3g_2^2) (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} + \overline{a_d^{pr}} (2 C_{\phi 2^2} a_d^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_d^{pr})) LF_{3,1,0}[m_q^p, m_d^r] - \\ & \frac{1}{2} g_1^2 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} + \overline{a_d^{pr}} (2 C_{\phi 2^2} a_d^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_d^{pr})) LF_{3,2,-1}[m_q^p, m_d^r] + \\ & \frac{1}{4} (-6 C_{\phi 2^2} \overline{a_d^{pr}} a_d^{pr} (7g_1^2 + 5g_2^2) - 21 \tilde{\mu} (g_1^2 + g_2^2) (C_{\phi 2\phi 1} \overline{y_d^{pr}} a_d^{pr} + \overline{C_{\phi 2\phi 1}} y_d^{pr} \overline{a_d^{pr}})) \\ & LF_{4,1,-1}[m_q^p, m_d^r] + \\ & (2 C_{\phi 2^2} \overline{a_d^{pr}} a_d^{pr} (3g_1^2 + g_2^2) + \tilde{\mu} (3g_1^2 + 2g_2^2) (C_{\phi 2\phi 1} \overline{y_d^{pr}} a_d^{pr} + \overline{C_{\phi 2\phi 1}} y_d^{pr} \overline{a_d^{pr}})) \\ & LF_{5,1,-2}[m_q^p, m_d^r] - 3 \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{pt} y_d^{sr} LF_{1,1,0}[m_q^p, m_q^s] - \\ & 6 C_{\phi 2^2} \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{pt} y_d^{sr} LF_{2,1,0}[m_q^p, m_q^s] + 6 C_{\phi 2^2} \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{pt} y_d^{sr} LF_{3,1,-1}[m_q^p, m_q^s] + \\ & \frac{1}{2} \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} (g_1^2 - 3g_2^2) LF_{2,1,0}[m_q^p, m_u^r] + \\ & \frac{1}{4} \tilde{\mu} ((g_1^2 - g_2^2) (C_{\phi 2\phi 1} \overline{y_u^{pr}} a_u^{pr} + \overline{C_{\phi 2\phi 1}} y_u^{pr} \overline{a_u^{pr}}) + 2 C_{\phi 2^2} \tilde{\mu} \overline{y_u^{pr}} y_u^{pr} (g_1^2 + g_2^2)) LF_{2,2,0}[\\ & m_q^p, m_u^r] - \frac{3}{2} \tilde{\mu} g_2^2 (\overline{C_{\phi 2\phi 1}} y_u^{pr} \overline{a_u^{pr}} + \overline{y_u^{pr}} (C_{\phi 2\phi 1} a_u^{pr} + 4 C_{\phi 2^2} \tilde{\mu} y_u^{pr})) LF_{3,1,0}[m_q^p, m_u^r] + \\ & \frac{1}{2} \tilde{\mu} g_2^2 (\overline{C_{\phi 2\phi 1}} y_u^{pr} \overline{a_u^{pr}} + \overline{y_u^{pr}} (C_{\phi 2\phi 1} a_u^{pr} + 4 C_{\phi 2^2} \tilde{\mu} y_u^{pr})) LF_{3,2,-1}[m_q^p, m_u^r] + \\ & \frac{3}{2} \tilde{\mu} g_2^2 (\overline{C_{\phi 2\phi 1}} y_u^{pr} \overline{a_u^{pr}} + \overline{y_u^{pr}} (C_{\phi 2\phi 1} a_u^{pr} + 4 C_{\phi 2^2}$$